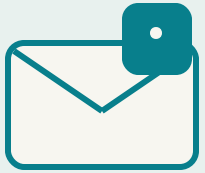
Why this design deserves credit.



Stratified randomisation

Helps balance sex and BMI across assigned groups.



Concealed allocation

Reduces scope for selection based on the next assignment.



Shared support

Makes unequal attention a less likely explanation.



Calibrated weighing

Less reliance on participants' own reports of their weight.

Advice did not guarantee equal intake.

HKD

Energy restriction
+ carbohydrate limit

−704

kcal/day

ERD

Energy restriction

−423

kcal/day

Calorie advice ≠ matched actual intake.

No controlled feeding ensured equal calorie intake.

Lim et al. 2024 · Methods 2.3, 2.5, 2.7 + Table 5 · Values are self-reported changes, not absolute intake.

Both composition and intake changed.



Diet composition

Carbohydrate restriction



Reported calories

Larger reduction in HKD

Both accompanied greater weight loss.

Their contributions were not experimentally isolated.

Lim et al. 2024 · Table 5 + Discussion · This comparison does not apportion the causal contribution of each change.

The mechanism is still uncertain.



Diet composition

Carbohydrate restriction



Reported calories

Larger reduction in HKD

A calorie-independent ketosis effect is unproven.

Eating less can be part of the programme’s benefit.

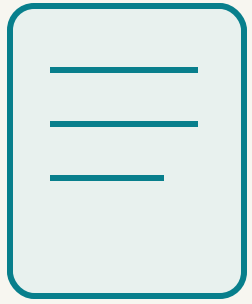
The two explanations may overlap; the trial did not isolate ketosis itself from energy-intake differences.

Was ketosis actually confirmed?

THE PRESCRIPTION

≤50 g

net carbohydrate/day



Food diaries



Ketosis?

A diary reports intake.

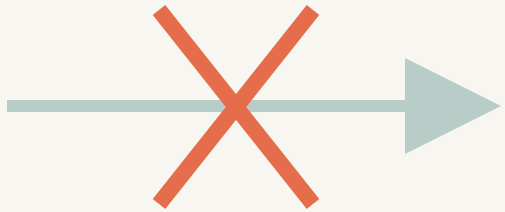
It does not measure a metabolic state.

Lim et al. 2024 · Methods 2.3, 2.7 + Discussion · Food diaries are self-reported.

Weight loss does not prove ketosis.



Weight loss was observed



Does not prove



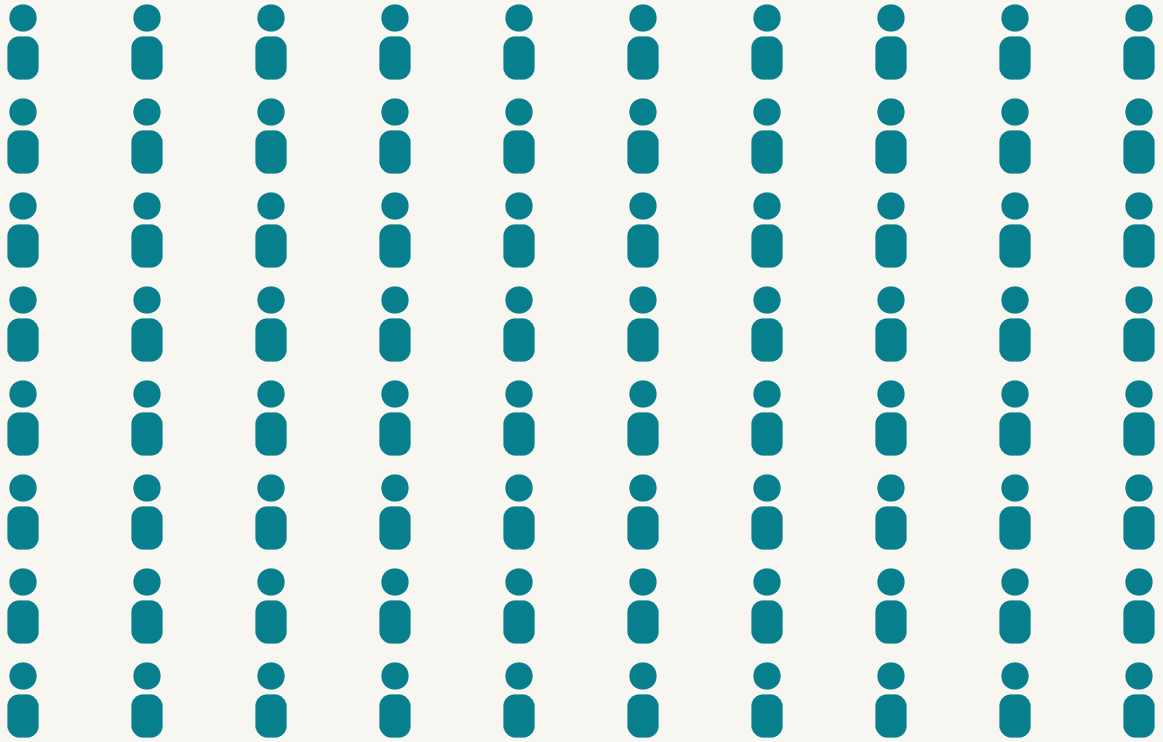
Confirmed ketosis

Calorie reduction can also produce weight loss.

The outcome does not identify a ketosis-specific effect.

Methods 2.3, 2.7 + Discussion · The trial tests assigned advice, not experimentally verified ketosis exposure.

Who stays in the picture?



Observed Same 80-person cohort

80

randomized at baseline

100% in the starting cohort

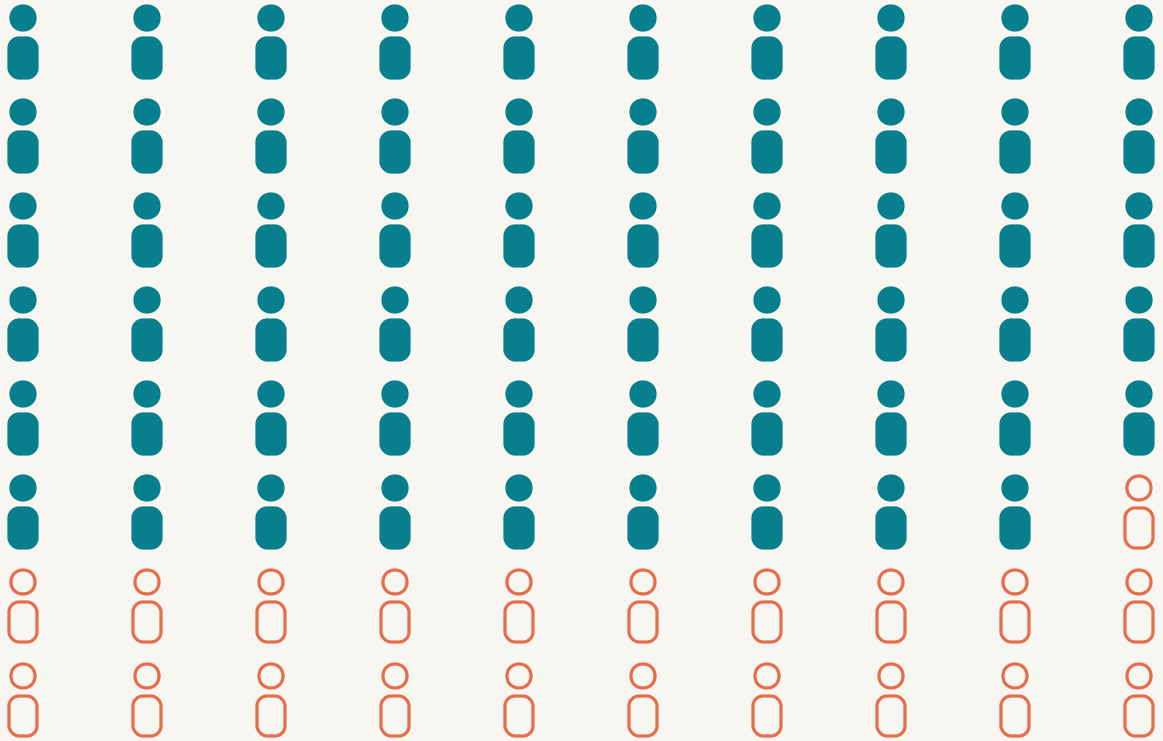
What happens next?

Follow the same cohort.

One figure = one participant

Lim et al. 2024 · Figure 1 + Table 2 · 6-month counts: HKD 29/41; ERD 30/39.

Who stays in the picture?



● Observed ○ Missing at 6 months

59

measured at 6 months

74% retained

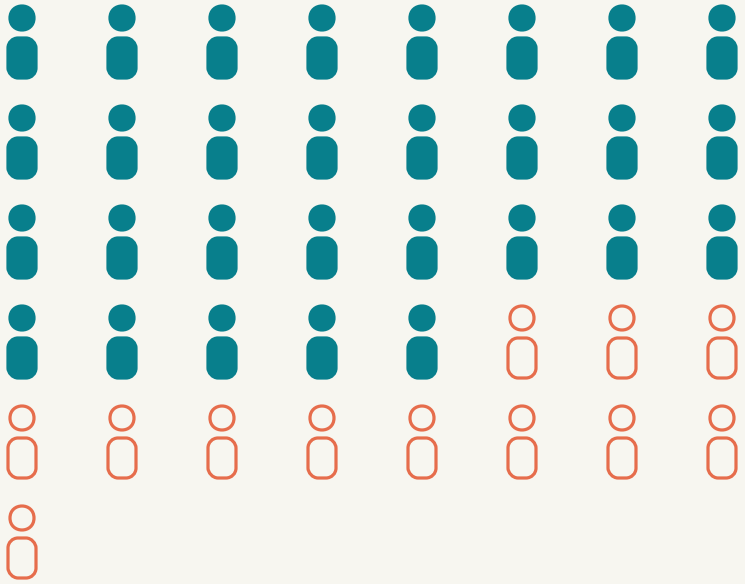
21 outcomes unknown

Missing weight ≠ failed diet.

HKD missing: 12 · ERD: 9

Lim et al. 2024 · Figure 1 + Table 2 · 6-month counts: HKD 29/41; ERD 30/39.

The missing weights could matter.



HKD: 29 measured + 12 missing

KNOWN

12 weights were not measured.

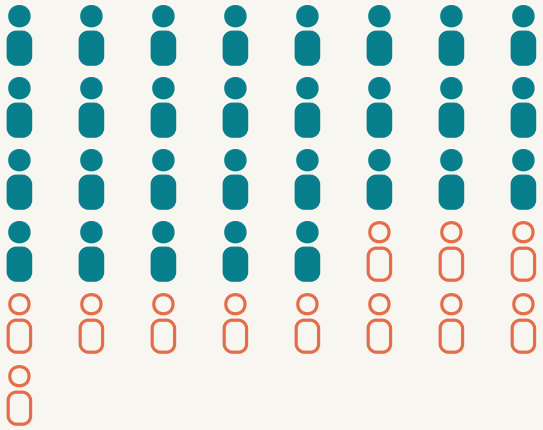
UNKNOWN

Less, similar or greater weight loss?

Their dietary adherence may be good.

Their unknown outcomes could change the group estimate.

Missing weights may bias the result.



29 measured
12 missing

IF POORER HKD RESPONDERS WERE MISSING…

The measured HKD subset would look more successful.

The apparent HKD advantage could be exaggerated.

Potential attrition bias; other missingness patterns could shrink the gap.

Missing-outcome handling is not clear in the report.

Figure 1 + Table 2 + Methods 2.9 · The actual direction of any missing-data bias cannot be established here.

Useful evidence, with clear limits.

WHY THE COMPARISON HAS VALUE

Better balance, fairer support, more objective weight data.

CALORIES	A different reported intake complicates attribution.
KETOSIS	The metabolic state was not objectively verified.
MISSING WEIGHTS	The measured subset may not represent everyone.

Lim SL et al. (2024), Nutrients 16:4380 · doi:10.3390/nu16244380